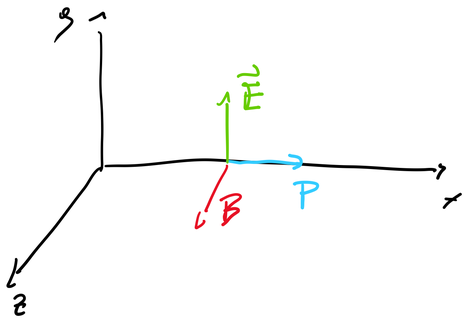
Intensità delle onde armoniche

2/5/21 1:29 PM



$$\vec{E} = \bar{E}_0 \operatorname{sen}(kx - \omega t) \qquad E_0 = c B_0 \qquad \left(\begin{array}{l} E_0 = v B_0 \\ \omega = k v \end{array} \right)$$

$$\bar{B} = \bar{B}_0 \operatorname{sen}(kx - \omega t) \qquad \omega = k c$$

$$\omega = \frac{2\pi}{T} = 2\pi \nu \qquad \lambda \nu = c \qquad \nu = \frac{c}{\lambda}$$

$$k = \frac{2\pi}{\lambda}$$

$$\omega = 2\pi \nu = 2\pi \frac{c}{\lambda} = k c$$

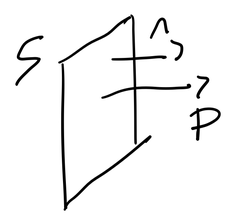
$$u = \frac{1}{2} \varepsilon E^2 + \frac{1}{2\mu} B^2 = \left(\frac{1}{2} \varepsilon E_0^2 + \frac{1}{2\mu} B_0^2 \right) \operatorname{sen}^2(kx - \omega t) = \varepsilon E_0^2 \operatorname{sen}^2(kx - \omega t)$$

$$\langle u \rangle = \varepsilon E_0^2 \langle \operatorname{sen}^2 \rangle = \frac{1}{2} \varepsilon E_0^2 = \frac{1}{2\mu} B_0^2$$

INTENSITÀ: $\qquad E \perp B$

$$I := |P| = \left| \frac{1}{\mu} E \times B \right| = \frac{1}{\mu_0} E_0 B_0 \operatorname{sen}^2(kx - \omega t)$$

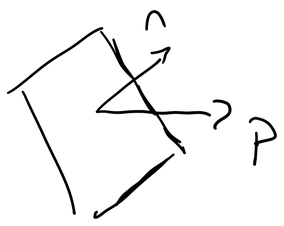
I: Flusso del vettore di poynting attraverso una superficie unitaria $\perp$ alla direzione di propagazione $\qquad P /\!/ n \Rightarrow P \cdot n = |P|$



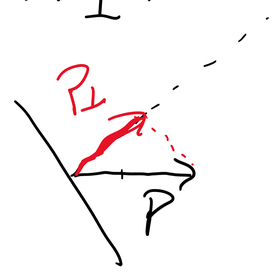
$$I = \iint \vec{P} \cdot \vec{n} \, dS = |P| \cdot |S| = |P|$$

NB:

Se $\vec{P}$ non $\perp$ S



$$I = |P_\perp|$$



INDICE di RIFRAZIONE:

$$n := \sqrt{\varepsilon_r \mu_r}$$

VELOCITÀ

$$v = \frac{c}{n} = \frac{1}{\sqrt{\varepsilon \mu}}$$

I media $\qquad E \perp B$

$$\langle I \rangle = \frac{1}{\mu} |E \times B| = \frac{1}{\mu} E_0 B_0 \langle \operatorname{sen}^2 \rangle = \frac{1}{2\mu} E_0 B_0 =$$

$$= \frac{1}{2\mu} \frac{E_0^2}{v} = \frac{\sqrt{\varepsilon \mu}}{2\mu} E_0^2 = \sqrt{\frac{\varepsilon}{\mu}} \cdot \frac{1}{2} E_0^2 = \frac{1}{2} \frac{\varepsilon E_0^2}{\sqrt{\varepsilon \mu}} = \frac{1}{2} \frac{c}{n} \varepsilon E_0^2$$

$$\boxed{= \frac{c}{n} \bar{u} = v \bar{u}}$$

PRESSIONE di RADIAZIONE

$$p_{em} = \frac{\bar{I}}{v}$$

Q moto

$$\bar{g} = \frac{\bar{P}}{v^2} = \varepsilon \bar{E} \times \bar{B}$$